



Differential motion attention network for efficient action recognition

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Abstract

Despite the great progresses achieved by commonly-used 3D CNNs and two-stream methods in action recognition, they cause heavy computational burden which are inefficient and even infeasible in real-world scenarios. In this paper, we propose differential motion attention network (DMANet) to specially highlight human dynamics toward efficient action recognition. First, we argue that consecutive frames contain redundant static features and construct a low computational unit for discriminative motion extraction to highlight the human action trajectories across consecutive frames. Second, as not all spatial regions in images play an equal role in depicting human actions, we propose an adaptive protocol to dynamically emphasize informative spatial regions. As an end-to-end lightweight framework, our DMANet outperforms costly 3D CNNs and two-stream methods by 2.3% with only $0.23\times$ computations and other efficient methods by 1.6% on Something–Something v1 dataset. Experimental results on two temporal-related datasets and the large-scale scene-related Kinetics-400 dataset prove the efficacy of DMANet. In-depth ablations further give both quantitative and qualitative support on its effects.

Keywords Action recognition · Temporal reasoning · Differential motion attention · Efficiency

1 Introduction

Past years have witnessed great success brought by deep convolutional neural networks (CNNs) in video action recognition. Temporal reasoning is one of the key factors in video understanding and has been explored by lots of methods [1–5] recently, which mainly fall into two groups. 3D CNNs [6–8] and their decomposed (2+1)D variants [9–11] propose to perform temporal reasoning in a straightforward way with cubic convolutions or consecutive spatial- and temporal-only convolutions. They turn out to be effective for modeling spatial–temporal dependencies but structurally consider each frame equally, leading to a heavy computation burden due to the dramatically increased parameters. Another common group is to employ a two-stream network [12], where one stream relies on RGB images for capturing static features and another stream adopts optical flow or temporal difference modality to model spatial–temporal dependencies. The prediction scores on action classes from both streams

are averaged in the end to perform classification to absorb complementary information from each other. Although two-stream network has a broad impact in action recognition, it only performs information fusion in the end of network while failing to capture movement patterns in its early or middle section, thus losing beneficial interactions between frames.

In this paper, we aim to build an effective architecture for *efficient* action recognition. Our intuition is derived from the observation in action recognition, human beings pay more attention to motion than static areas. Considering consecutive frames always contain similar and redundant spatial features which make it hard to capture the movement of the actor during a local period, motion feature could more clearly show the movements of actors and thus serve as a better choice to represent actions. Thus, we propose to employ motion features by temporal difference operators, enhancing the ability of model to perceive regional movements across consecutive frames. To save computations and achieve compact spatial modeling, we decouple feature extraction into spatial and temporal groups in parallel, corresponding to independent spatial modeling and temporal difference modeling. Our attempts consist of two parts.

First, we employ temporal difference operators to build powerful and low computational temporal reasoning units. Previous works have revealed that consecutive frames always

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own *similar* contexts [13, 14]. Repeatedly computing similar spatial features for these frames may lead to redundant representations and fail to fully capture the temporal movements of the actor. We incorporate temporal difference operators to redefine shift operations to build a low-computation module for capturing local motion. By combining temporal difference operators with shift operations, our model avoids the redundant extraction of these similar features and emphasize short-term motion features.

Second, temporal difference operators not only provide discriminative motion features, but also serve as an indicator for informative spatial features. Previous methods such as 3D CNNs and their variants treat frames in a spatial-temporal neighborhood equally and employ cubic convolutions to model spatial-temporal dependencies. We argue that they may introduce redundant spatial-wise features. In contrast, we employ motion features to highlight the movements of human beings and emphasize these regions to distinguish actions. Inspired by this, we propose a differential motion module in this paper to recognize informative spatial features through temporal difference operators. Specially, motion features act like attention maps, guiding models to pay attention to informative changing regions. Hence, models get less distracted by background noise or identical scenes which may provide confused or redundant information.

In this paper, we present a lightweight framework, DMANet, toward *efficient* action recognition. To verify the efficacy of DMANet, we implement it with a popular backbone ResNet-50 and evaluate it on two challenging temporal-related datasets and a large-scale scene-related dataset Kinetics-400 [7]. The experiment results on Something-Something v1 and v2 datasets [15] show that DMANet is able to outperform costly 3D CNNs and two-stream methods by 2.3% with only $0.23\times$ computations as well as other efficient methods. Experimental results on two temporal-related datasets and the large-scale scene-related Kinetics-400 dataset demonstrate its efficacy.

2 Related works

2.1 Spatial-temporal modeling in action recognition

Before CNNs meet action recognition, most methods rely on hand-crafted features for recognition. For example, IDT [16] aims to capture motion regions and STIP [17] constructs space-time interest points. However, they fail to flexibly extract high-dimensional features, while CNNs make up this drawback. Spatial-temporal modeling is one of the key factors in action recognition, and various methods have been developed, which can be divided into following groups: 1. Employing 3D convolution [6–8] or its (2+1)D variants

[9–11]. To mitigate beneficial experience from image recognition and deal with videos, researchers propose to inflate 2D models and employ 3D convolutions to tackle cubic input. However, direct inflation leads to a heavy computation burden. To relieve this limitation, some methods decompose 3D convolution into (2+1)D [9–11], achieving better performance meanwhile. 2. Two-stream networks [12, 18–20]. One stream relies on RGB images to capture context information, while another stream often employs optical flow for modeling cross-frame features. Two-stream networks turn to be effective but consume twice the computations. Besides, optical flow has to be computed offline in advance. To relieve this issue, some methods [21, 22] try to generate optical flow from RGB inputs and perform classification in a unified framework, but yield lower performance. 3. Transformer-based methods [23–25]. These methods usually employ factorized spatial and temporal transformers or directly use a spatial-temporal transformer to build local spatial-temporal relationships. As the parameters and FLOPs of transformers are usually larger than 2D CNNs, transformer-based methods in video understanding usually own larger computational overheads than 2D-CNN-based methods. 4. Partially adopting 3D convolutions [13, 26, 27]. Some methods incorporate 3D convolutions to capture long-range dependencies under limited cost. For example, GST [27] and STM [13] inflate a portion of convolutions into 3D upon 2D backbones. With few additional parameters and computations, they are able to achieve comparative results with 3D methods. 5. Networks with shift operation toward efficient recognition citesudhakaran2020gate, TEINet, shao2020temporal. As the pioneer in action recognition, TSM [28] introduces shift operation for capturing cross-time features. GSM [29], TEINet [30] and TIN [31] make a step further with more powerful operators such as gated control, learnable shift and deformable shift. Others [13, 14, 30, 32–34] progress by designing more powerful operators. We take a step further by capturing discriminative motion features. Our work also follows this line and achieves superior performance than all above.

2.2 Motion perception in video understanding

Motion perception is a key factor in video understanding. To well capture the short-term and long-term motion information in videos, various kinds of methods have been proposed from different perspectives. Su et al. [26] present a sequential approach for articulated motion estimation from a 2D skeleton sequence. Lee et al. [35] propose long-term motion context memory with memory alignment learning and memory query decomposition to capture long-term motion. In this paper, we focus on the usage temporal difference operators, which serves as a cheap and effective operation to capture motion features. Compared to optical flow, temporal difference can be computed online and does not need huge

memory to store in advance. However, it owns less powerful ability in modeling region changes. Some methods incorporate temporal difference operators inside the network to capture motion features. For example, STM [13] and TEINet [30] employ temporal difference to perform temporal modeling, while TEA [14] adopt temporal difference for channel weighting. MotionFormer [36] designs trajectory attention to capture the movements of actions based on temporal difference. Kim B et al. [37] designs a motion-based transformer to handle the input variations in action recognition. However, they simply perform subtraction between features of adjacent frames, which may not fully realize the potential of temporal difference. We employ temporal difference operators both temporally and spatially, to model long-term dependencies and emphasize informative regions complementarily.

2.3 Attention mechanism in video understanding

Since the introduction of Transformer series, the attention mechanism has been broadly employed in video understanding. Action-net [38] designs a spatiotemporal, channel and motion excitation module to emphasize the key information from different aspects. VDRAN proposes [39] a video disentangling module to decompose segment-wise video into motion cues and object cues. Li et al. [40] introduce a multi-branch multi-scale graph convolution network to build the motion information in 3D skeletons. COSNet [41] introduces global co-attention mechanism to emphasize the importance of inherent correlation among video frames. HiGraph [3] employs graph kernels to build a space-time correspondence framework to model relations between different instances in videos. TEA [14] propose a temporal excitation and aggregation network to model short-term motion information in videos. Our method builds an efficient multi-scale architecture to perceive the local spatial-temporal features, which achieve better results than previous methods based on expensive 3D CNNs.

3 Differential motion attention network

In this section, we give a detailed description of our DMANet. First, we give the notations in this paper. Then, we incrementally offer the details of two building blocks of DMANet to perform temporal reasoning and differential motion perception. Our temporal difference module aims to capture distinct motion features, avoiding redundant spatial information. Next, our spatial weighting module emphasizes the informative areas in accordance with region changes in a data-dependent way. Equipped with efficient temporal difference operators, our modules only consume small additional computation costs. Finally, we provide the implementation details to instantiate DMANet with a ResNet-50 backbone.

3.1 Preliminaries

Formally, input videos always have various lengths and a natural choice is to divide them into T segments with equal durations and no overlapping. Then, a frame is randomly picked from each segment and all frames are concatenated in turn. Thus, each video can be represented as a sequence of frames $\{X^1, \dots, X^T\}$ with total size $R^{C \times T \times H \times W}$. Here, C represent channel numbers, while H and W denote height and width alternatively. Frames are then processed by a shared 2D backbone where their prediction scores are averaged in the end to perform classification.

3.2 Temporal difference module

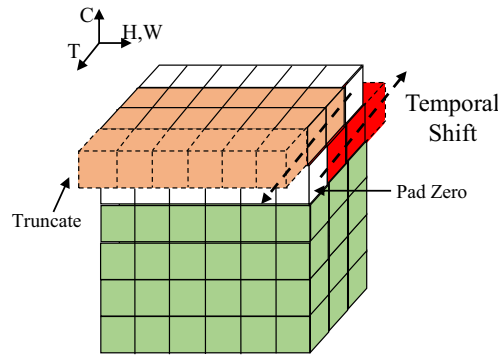
3.2.1 Revisiting TSM

Although the architecture described above consumes few frames for prediction, a drawback appears that cross-frame features only interlace at the end of network. To tackle this limitation, TSM [28] proposes lightweight shift operations. As illustrated in Fig. 1a, features in orange and red color are shifted backward and forward separately, capturing bidirectional cross-time features from adjacent frames. Compared to normal design, shift operations introduce *no* additional parameters and can be flexibly incorporated into network blocks to capture *all-stage* long-term dependencies. They offer a new way for temporal reasoning and are broadly adopted by following methods [29–31, 42].

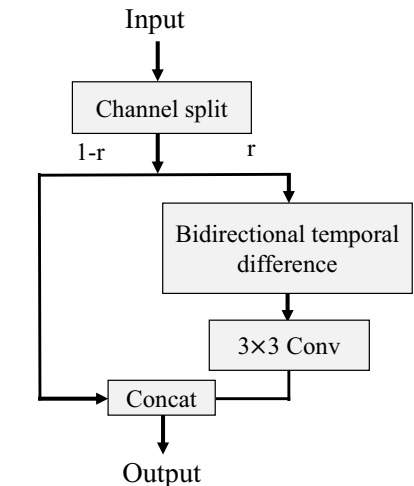
3.2.2 Temporal reasoning

Although TSM could increase the temporal receptive field by shifting spatial features along the temporal dimension forward and backward, we argue that naively shifted features are not powerful enough for distinguishing actions. Considering consecutive frames always contain *similar* and *redundant* spatial features, shifted features usually own similar context with features of the current frame and thus make it difficult for models to distinguish the movements of the actor. To tackle this issue, we introduce motion information into the shift operators to emphasize the movements of actors. Considering that in a local temporal duration, one single frame could already provide enough spatial information. Introducing motion features could clearly reveal the trajectories of actors across consecutive frames, avoiding paying much attention to meaningless information like backgrounds. Thus, we attempt to incorporate lightweight temporal difference operators into shifted features to highlight the short-term motion.

Fig. 1 **a** Illustration of temporal shift operation. It is built computationally free on top of a 2D convolution, but can offer strong temporal modeling power. **d** Illustration of our temporal difference module. We incorporate temporal difference operators into shift operations to highlight action dynamics. Best viewed with color



(a) Temporal Shift Operation



(b) Temporal Difference Module

Typically, temporal shift operations are implemented as:

$$X^{out} = \text{Concat} \left\{ X_{t-1}^{\frac{r}{2}C}, X_{t+1}^{\frac{r}{2}C}, X_t^{(1-r)C} \right\} \quad (1)$$

where r denotes the channel shift ratio and is set as $1/4$ according to TSM ($1/8$ in each direction). C is the number of input channels. To demonstrate cross-frame motion, we incorporate temporal difference operators to enhance shift operations as:

$$X^{out} = \text{Concat} \left\{ X_t^{\frac{r}{2}C} - X_{t-1}^{\frac{r}{2}C}, X_{t+1}^{\frac{r}{2}C} - X_t^{\frac{r}{2}C}, X_t^{(1-r)C} \right\} \quad (2)$$

where the current frame's features are preserved to offer static information and motion features between consecutive frames are emphasized, avoiding spatial redundancy.

A factor matters that videos often suffer from interference like camera motion, light transformations and background noise. Temporal difference between consecutive frames eliminates identical backgrounds but also *amplifies* pixel disturbance. We add a transformation module behind to smooth these disturbances. It is instantiated as a 3×3 convolution for filter as default.

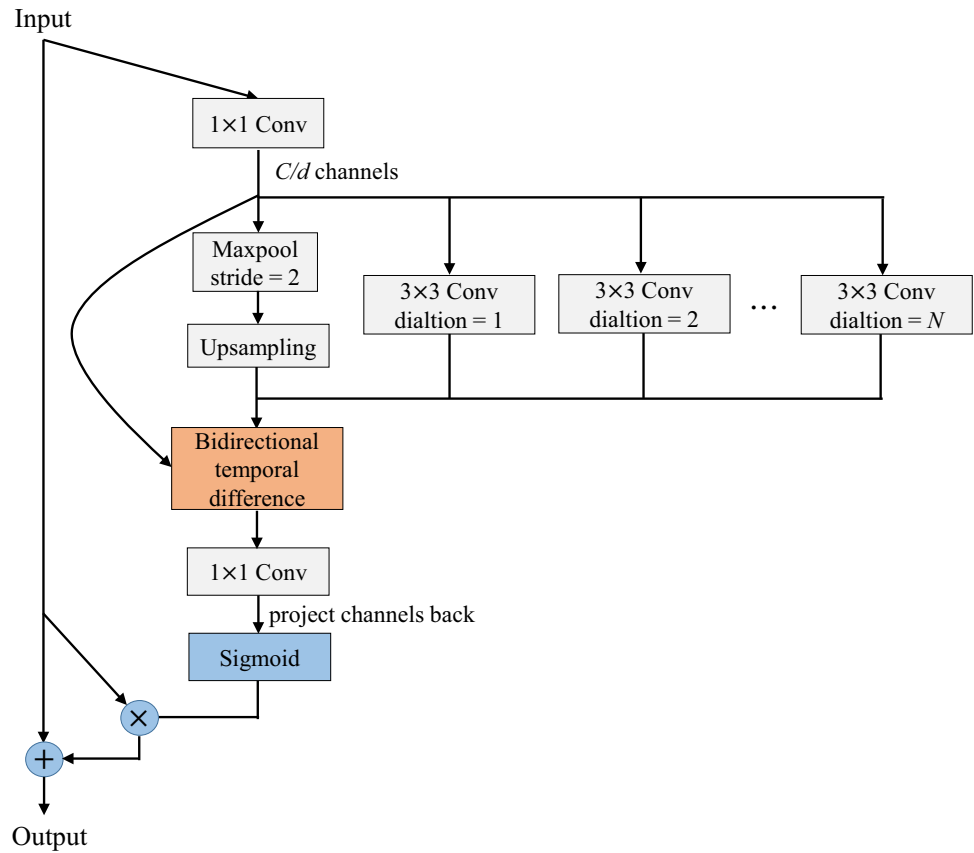
3.3 Differential motion module

Temporal difference operators not only provide discriminative motion features, but also serve as an excellent *indicator* for informative *spatial* regions. While most existing methods employ convolution kernels to operate pixels in a sliding window, they treat different regions *identically*. We argue it is *inappropriate* to treat pixels equally and some particular areas deserve more attention. Formally, static information such as background, non-interacted objects and still body

parts is broadly distributed in frames and one frame provides enough context. Motion between consecutive frames depicts the local movement pattern of actions and plays a vital role in modeling local human dynamics. Thus, motion regions need more attention while common spatial areas deserve less. We propose to employ temporal difference operators to adaptively strengthen informative regions while compressing others, resulting in a differential motion module.

As illustrated in Fig. 2, we first aggregate multi-scale features and then perform differential motion perception. Multi-scale features have been proved to be effective in various fields [43–52] but few explored in action recognition. The misalignment of spatial locations in consecutive frames is an important concern for capturing motion. To smooth the difference between consecutive frames, we devise a multi-scale architecture for capturing long-range spatial features. Inspired by dilated convolution [53], we adopt dilated convolutions with multiple dilation rates to obtain long-range spatial features under limited costs. Dilated convolutions allow larger receptive field without growing the size of convolutional kernels. We first reduce channels by a factor of d with a 1×1 convolution to save computations. Then, we employ N convolution branches and one max pooling branch in parallel to capture long-range spatial features. All convolutions own the same kernel size 3 with increased dilation rates in $\{1, 2, \dots, N\}$. With dilated convolution, we can obtain robust represents from different receptive fields with maximum receptive field up to $2N + 1$, under limited computations. We employ grouped convolutions for efficiency. The max pooling branch aims to capture compact representations in smaller features maps for locating motion regions, which is followed by an upsampling layer. Outputs from different branches are averaged to obtain robust multi-scale features

Fig. 2 Illustration of the differential motion module. We devise a multi-scale mechanism to tackle misalignment of spatial locations between consecutive frames to emphasize informative regions. Outputs from different branches are averaged to perform motion attention perception



as:

$$\hat{X}_t = \frac{1}{N} \sum_{i=0}^N \text{Conv}_i(X_t, n=i) \quad (3)$$

where $\hat{X}_t$ represents the fused output features and n denotes the dilation rate.

Then, we incorporate temporal difference operators to compute bidirectional motion features F_t with $\hat{X}_t$ as:

$$F_t = \text{Concat}([\hat{X}_{t+1} - X_t; \hat{X}_t - X_{t-1}]). \quad (4)$$

Here, F_t depicts the local movement patterns between consecutive frames.

Given F_t , a 1×1 convolution is employed to project channels back. We then pass F_t through a sigmoid function to compute spatial weights in $[0,1]$. Finally, we employ these spatial weights to enhance frame-level features and incorporate them mainstream via a residual connection as follows:

$$X_t = X_t + \text{Sigmoid}(\text{Conv}_{1 \times 1}(F_t)) \quad (5)$$

3.3.1 Discussion

Attention mechanism has been commonly visited in various fields [54–60] for long-range spatial aggregation or channel weighting. While serving as attention maps for differential motion perception, temporal difference is a unique factor in temporal-related tasks. As shown in our experiments, temporal difference owns an advantage over attention operation and they are complementary in design.

3.4 Model architecture

Given the details of temporal difference module and differential motion module, we introduce how to instantiate DMANet with a ResNet-50 backbone. As illustrated in Fig. 3, the ResNet-50 backbone is consisted of a Conv1 layer and four blocks (Res1-Res4), attached by a linear layer for classification. Inside each block, we insert a differential motion module before Conv2 for spatial aggregation. Then, we place a temporal difference module after Conv3 to model human dynamics. These modules can be built on top of 2D backbone blocks painlessly and flexibly. After extracting spatial features with all four blocks, we average the features of all input frames and send them into a linear layer to perform classifica-

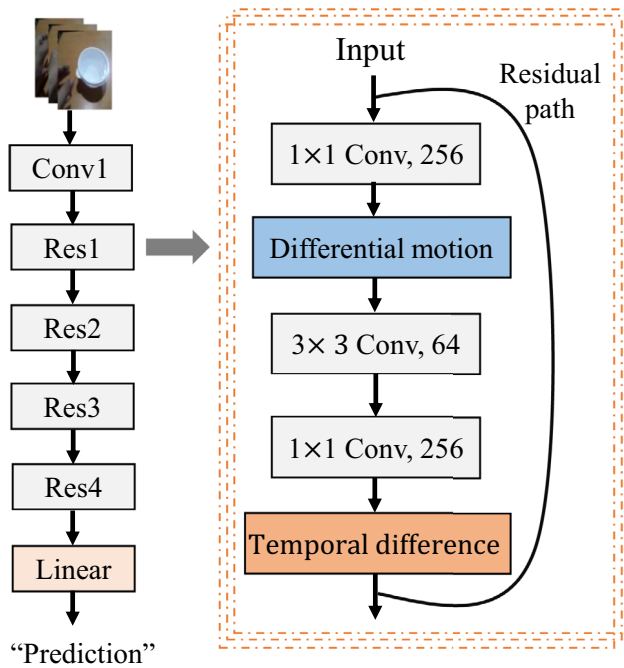


Fig. 3 Our DMANet instantiated with ResNet-50 as backbone. An example of Res1 block is illustrated in the right

tion. The training procedure is supervised by a cross-entropy loss.

3.5 Model complexity

The advantage of traditional 3D CNN is to effectively model local spatial-temporal information. However, the usage of expensive 3D convolutions brings heavy computational burdens. For example, compared to R2D, I3D consumes around five times of computations (153 GFLOPs v.s. 33 GFLOPs), which may be unfeasible in real-life scenarios. To handle this issue, we build our DMANet upon the lightweight 2D backbone and propose to capture local spatial-temporal information with efficient temporal difference operators and differential motion perception. The introduced operators only consume quite few computations (2.1 GFLOPs). Overall, compared to I3D with 153 GFLOPs, our DMANet only consume 35.1 GFLOPs, but achieves better performance across a wide range of datasets especially motion-focused datasets like Something–Something [15].

4 Experiments

4.1 Datasets

Kinetics-400 [7]: Kinetics-400 is a large-scale dataset containing about 240k training videos and around 20k validation videos collected from YouTube over 400 classes. It is more

Table 1 Ablations of DMANet on Something–Something v1 dataset [15]

Model configurations	FLOPs	Accuracy (%)
Temporal difference module		
Baseline	33.0G	48.5
+ temporal difference	33.0G	49.2
+ background noise relieve	34.0G	49.7
Differential motion module		
Multi-scale feature extraction		
No Conv	34.0G	49.7
7 × 7 grouped Conv	37.1G	50.2
dilations = [1]	34.3G	50.3
dilations = [1,2]	34.7G	50.6
dilations = [1,2,3]	35.1G	50.9
dilations = [1,2,3,4]	35.6G	51.2
dilations = [1,2,3] + maxpool	35.1G	51.5
Compressed ratio d		
$d = 4$	36.1G	51.4
$d = 8$	35.5G	51.2
$d = 16$	35.1G	51.5
Fusion method		
$X_t^{out} = w \times X_t$	35.1G	49.8
$X_t^{out} = X_t + w \times X_t$	35.1G	51.5

We give ablations of temporal difference module and differential motion module alternatively

Table 2 Comparison with other attention-based modules on Something–Something v1 dataset [15]

Methods	FLOPs	Boost (%)
Temporal difference module		
+Differential Motion Module	35.1G	1.8
+Non-local Module	159.1G	0.8
+SENet	34.6G	0.5
+CBAM	35.3G	1.0
+Differential Motion & CBAM	36.4G	2.4

scene-related and mainly composed of daily activities highly correlated with fine-grained context features.

Something–Something [15]: Something–Something is a large-scale temporal-related dataset with special emphasis on motion. It performs action with different objects and similar backgrounds, aiming to draw attention to actions. It needs careful attention on details which may confuse the model, for example, pretending to pick something up and picking something up need special attention on behaviors to classify. The first version of Something–Something contains about 11k clips over 174 labels, while the second version is much bigger containing around 22k clips. We train our model on the training set and report accuracies on the validation set.

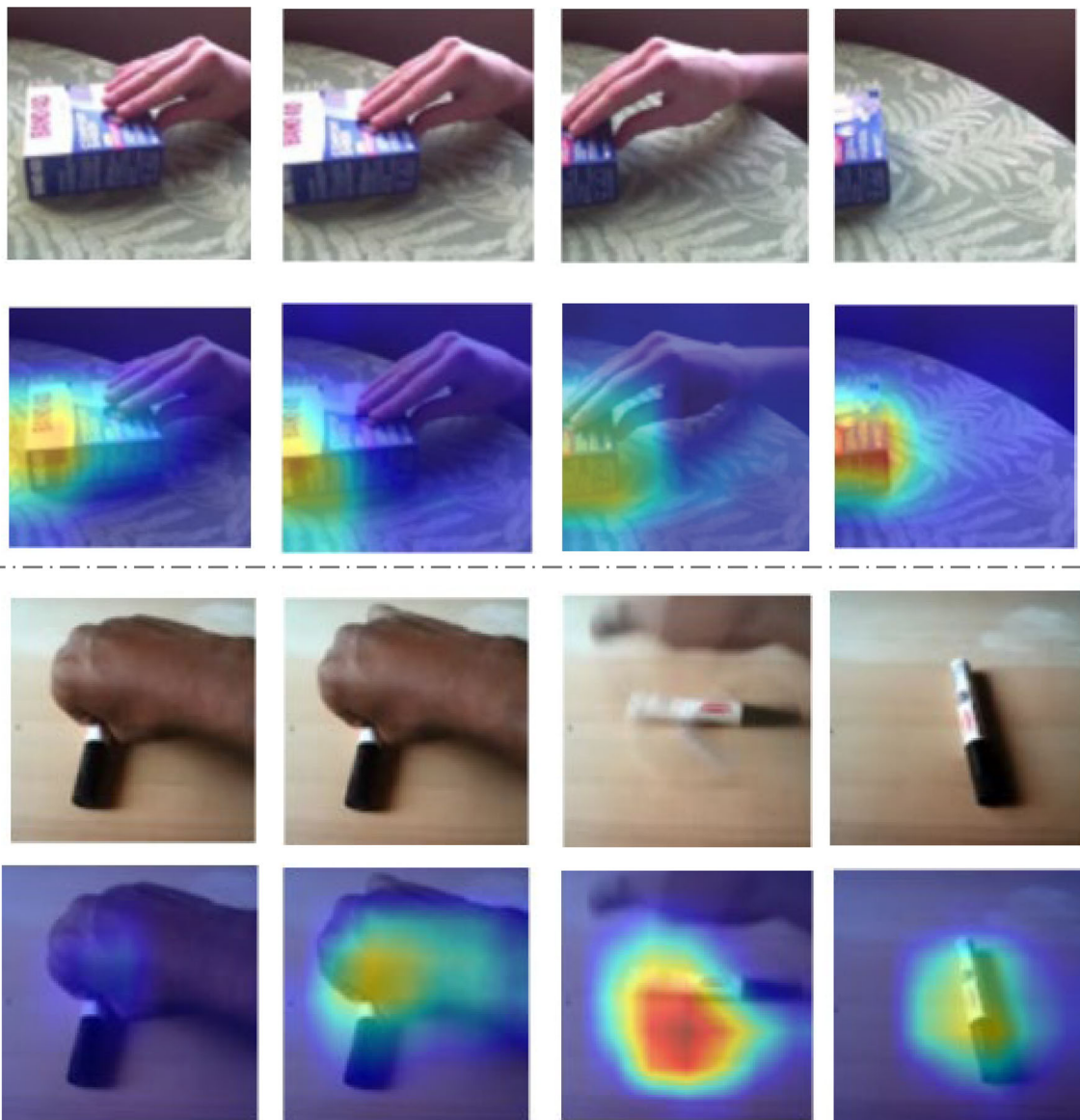


Fig. 4 Visualization of class activation maps with CAM [63]. Top rows: original images. Bottom: class activation maps

4.2 Training details

Following TSM, we present a uniform sampling strategy to obtain $\{8, 16\}$ frames with equal durations and no overlapping. We train our model with initial learning rate as 0.02 and adjust it with cosine learning rate scheduler with total 60 epochs. Our model is trained on a 4-2080ti machine with batch size 32 and weight decay $5e-4$ in PyTorch [61]. We adopt ResNet-50 [62] as our backbone to extract spatial features for each frame. The features of different frames are averaged and sent into a fully connected layer to perform prediction. The training process is supervised by a cross-entropy loss. During training, input images are first cropped by the shorted side into $[256, 320]$ and then crops with size

$[224, 224]$ are randomly selected. During testing, images are first resized by the shorted side to 256 and then center crops are adopted. For Kinetics-400 dataset [7], the training setting is a little different. The total epoch is 120. Besides, following NL-Net [54], we try a 10-clip and 3-crop sampling scheme where three crops of $[256, 256]$ and 10 clips are used for testing.

4.3 Ablation study

In this section, we incrementally build up our model to demonstrate the effects of components in DMANet on Something–Something v1 dataset [15].

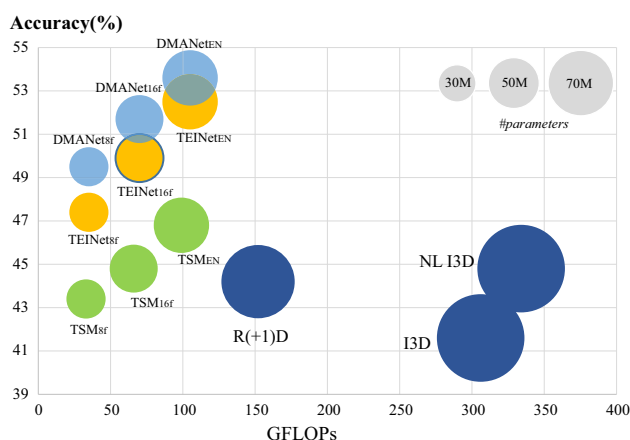


Fig. 5 Accuracy-FLOPs trade-off of popular models in video recognition. The area of circles indicates the parameters of corresponding model

4.3.1 Temporal difference module

We give ablations of temporal difference module as listed in the upper part of Table 1. When we incorporate temporal difference operators into network design, we observe 0.7% performance boost which proves the effect of motion features against naively shifted features for distinguishing actions. It demonstrates that models benefit more from discriminative motion features than naively shifted features. Practically, motion features clearly point out the trajectories among frames. After we place a transformation module behind to relieve noise, 0.5% performance boost is witnessed. Totally, incorporating temporal difference operators with shift operations greatly enhances temporal modeling ability with small computation overhead (1.0 GFLOPs).

4.3.2 Differential motion module

Then, we offer ablations of the differential motion module. We give ablations of differential motion module as listed in the lower part of Table 1. We incrementally add multiple convolutions with different dilation rates to testify the effectiveness of multi-scale features. As observed, accuracy keeps increasing when k grows and starts pacing slowly when $k = 3$. It demonstrates larger and multi-scale dilations help relieve the misalignment of spatial locations between consecutive frames but receive marginal benefit from distant pixels. Dilation rate N is set as 3 as default for accuracy-computation trade-off. Compared to normal design (No Conv in Table 1), a single grouped convolution kernel employed for spatial weighting improves 0.6% accuracy which demonstrates the effectiveness of differential motion mechanism with temporal difference. We also employ a single 7×7 grouped convolution kernel for comparison which matches the maximum receptive field of our differential motion module. While

a single 7×7 grouped convolution kernel owns 2.0 GFLOPs more than our dilated design, it still behaves 1.3% below which proves the efficiency of multi-scale feature extraction. We then add the max pooling branch and witness 0.2% performance boost. Finally, we testify the choices of channel compression ratio d and the fusion method. We offer d three choices in $\{4, 8, 16\}$ and observe similar performance. Considering efficiency, we set d as 16 as default. As for the fusion method, we try to incorporate differential motion mainstream by direct weighting or via a residual connection and observe the latter performs better. We analyze that direct weighting may hurt the ability of differential motion, while residual connections complement it with motion features.

4.3.3 Comparison with attention-based modules

We compare our differential motion module with several popular attention-based modules in Table 2. We take Non-local Net [54], SENet [55] and CBAM [56] as examples due to their extraordinary performance. Non-local Modules bring 0.8% performance boost but own a much heavier computation burden. SENet and CBAM alternatively employ channel attention or channel and spatial attention to model feature relationships and gain 0.5%/1.0% accuracy promotion. Our differential motion module makes 1.2% progress and surpasses them all via unique temporal difference in temporal-related tasks. We also find that differential motion with temporal difference is complementary to attention-based modules. Combining differential motion module and CBAM gains 1.6% performance boost.

4.3.4 Visualizations

To verify the effectiveness of our differential motion module, we select two actions from the Something–Something v1 dataset as examples and draw the class activation maps with CAM [63] in Fig. 4. Our differential motion module acts like attention maps and pays more attention to action-related regions while suppressing less beneficial features. Notably, the box, hands and pen which are highly correlated with actions are highlighted by our DMANet. Features of background area and non-related objects are suppressed. It is also worth noting that for the same object, e.g., pen in the bottom sub-figure, it is highlighted when pushed by hands but paid less attention in the motionless state. This demonstrates our DMANet is able to adaptively pay attention to highly correlated objects in specific frames (Tables 3, 4).

4.3.5 Comparison with other temporal modules

We compare other DMANet with other efficient temporal-related methods in Table 5. With 8-frame input and the same backbone (ResNet-50), all methods own similar com-

Table 3 Comparisons of DMANet against other state-of-the-art methods on the validation set of Something–Something v1

Methods	Backbone	#Frames	GFLOPs	Top1 (%)	Top5 (%)
TSN [19]	ResNet-50	8	33 G	16.8	–
TRN [64]	BN-Inception	8	16 G	34.4	–
I3D+GCN [65]	ResNet-50	32	158 G	43.4	75.1
I3D [7]	ResNet-50	32	153 G	41.6	72.2
R(2+1)D [9]	ResNet-34	32	152 G	45.7	–
S3D [10]	BN-Inception	64	66 G	48.2	78.7
TSM [28]	ResNet-50	8	33 G	45.6	73.2
TSM [28]	ResNet-50	16	65 G	47.2	74.5
TSM [28]	ResNet-50	8+16	98 G	49.7	79.0
TEINet [30]	ResNet-50	8	35 G	47.4	–
TEINet [30]	ResNet-50	16	66 G	49.9	–
TEINet [30]	ResNet-50	8+16	99 G	52.5	–
GST [27]	ResNet-50	16	59 G	48.6	76.6
TIN [31]	ResNet-50	8+16	101 G	49.6	78.3
STM [13]	ResNet-50	16	67 G	49.8	–
TEA [14]	ResNet-50	16	70 G	51.9	80.3
ABM [66]	ResNet-50	16×3	35 G×3	46.1	–
TrajectoryNet [67]	ResNet-18	7×10	–	47.8	–
CorrNet [68]	ResNet-50	32×10	115 G×10	49.3	–
V4D [69]	ResNet-50	8×4	168 G	50.4	–
SmallBigNet [70]	ResNet-50	8+16	157 G	50.4	80.5
TANet [71]	ResNet-50	8+16	99 G	63.6	79.3
AMS-Net [33]	ResNet-50	16	94 G	55.3	82.4
ARP [34]	ResNet-50	16	72 G	53.8	82.5
DMANet (Ours)	ResNet-50	8	35 G	51.5	78.7
DMANet (Ours)	ResNet-50	16	70 G	53.8	80.8
DMANet (Ours)	ResNet-50	8+16	105 G	55.7	82.1

Table 4 Comparisons of DMANet against other state-of-the-art methods on the validation set of Something–Something v2

Methods	#Frames	Backbone	Val (Top1)	Val (Top5)
TSM [28]	8+16	ResNet-50	63.5	88.6
TEINet [30]	8+16	ResNet-50	65.5	89.9
TRN [64]	8+8	BN-Inception	55.5	83.1
TIN [31]	8	ResNet-50	60.0	85.5
GST [27]	16	ResNet-50	62.6	87.9
STM [13]	16	ResNet-50	64.2	89.8
SmallBigNet [70]	8+16	ResNet-50	63.3	88.8
NUTA [72]	32×2	I3D-101	98 G	63.0
SIFAR [73]	16	SIFAR-L-14	64.2	88.4
TAda [74]	32	ConvNeXt-T	67.1	90.4
AMS-Net [33]	8+16	ResNet-50	67.0	90.9
ARP [34]	8+16	ResNet-50	68.0	91.7
DMANet (Ours)	8	ResNet-50	65.2	88.1
DMANet (Ours)	16	ResNet-50	66.7	89.0
DMANet (Ours)	8+16	ResNet-50	68.5	89.7

Table 5 Comparison with other temporal modules with 8-frame input on Something–Something v1 dataset [15]

Methods	FLOPs	Accuracy (%)
TSM [28]	33.0G	45.6
GST [27]	29.5G	47.0
TIN [31]	34.0G	47.0
TEINet [30]	35.0G	47.4
TEA [14]	35.0G	48.9
DMANet	35.1G	51.5

putations with our DMANet. We find the performance of DMANet is much better than these competitors under similar cost, owing to cheap but powerful temporal difference operators.

4.3.6 Model complexities

To show the accuracy-FLOPs trade-off of popular models in video recognition, we visualize them in Fig. 5. The accuracy is tested on the validation set of Something–Something v1 dataset [15], and the area of circles indicates the parameters of corresponding models. We select TSM [28] and TEINet [30] as examples of 2D models which own similar computations with our model. Compared to 3D model family, 2D models are in the upper left of the figure which are of fewer costs and achieve similar or better results. TSM models long-range dependencies through efficient shift operations, and TEINet promotes it by enhancing the motion-related features. Con-

sidering that TSM and TEINet are already designed to be both efficient and effective, DMANet achieves more competitive results than them with similar computation costs.

4.4 Comparisons with state of the arts

To demonstrate the effectiveness of DMANet on modeling long-range dependencies, we compare DMANet with other state-of-the-art methods on two temporal-related datasets Something–Something v1 [15] and v2 [15]. The results are reported in Tables 3 and 4, respectively. We implement DMANet with 8-frame or 16-frame input in practice and observe 16-frame DMANet consistently achieves better results. Compared to TSN [19] and TRN [64] with only late temporal fusion, our DMANet achieves significant progress owing to all-stage temporal reasoning. Compared to 3D CNNs [7] and their (2+1)D variants [9, 10], our DMANet surpasses them by a large margin with $0.23\times$ computations. Although they straightforwardly model spatial–temporal dependencies with cubic convolutions, we argue they generate redundant spatial–temporal features. Specially, compared to I3D+GCN [65] which explicitly locates objects by R-CNN [77] with heavy computation cost, our DMANet outperforms it by cheap temporal difference operators in a data-dependent way. Finally, we compare our DMANet with other specially design methods [13, 14, 27, 29–31] toward efficient action recognition. Our DMANet consistently outperforms them on both datasets under similar budgets and achieves new state of the arts (Table 5).

Table 6 Comparisons of DMANet against other state-of-the-art methods on the Kinetics-400 dataset

Methods	Frames×Crops×Clips	FLOPs	Top1 (%)	Top5 (%)
TSN-RGB [19]	25×1×10	80G×1×10	72.5	90.2
STM-RGB [13]	16×3×10	67G×3×10	73.7	91.6
TSM-RGB [28]	16×3×10	65G×3×10	74.7	–
TEA [14]	16×3×10	70G×3×10	76.1	92.5
TEINet [30]	16×3×10	66G×3×10	76.2	92.5
R(2+1)D [9]	32×1×10	152G×1×10	74.3	92.1
SmallBigNet [70]	8×10×3	57G×4×3	76.3	92.5
NL I3D-RGB [54]	128×3×10	282G×3×10	76.5	93.3
ip-CSN [75]	32×3×10	83G×30	76.7	92.3
SlowFast [18]	(8+32)×3×10	66G×3×10	77.0	92.6
CorrNet [68]	32×1×10	115G×1×10	77.2	–
NUTA [72]	64×1×10	98G×1×10	76.8	–
LgNet [76]	64×1×10	63G×1×10	77.6	93.0
ARP [34]	16×3×10	72G×3×10	77.1	93.4
AMS-Net [33]	16×3×10	137G×3×10	78.0	93.3
DMANet (Ours)	8×3×10	35G×3×10	76.4	92.1
DMANet (Ours)	16×3×10	70G×3×10	77.5	92.9
DMANet (Ours)	(8+16)×3×10	105G×3×10	78.5	93.8

We then conduct experiments on Kinetics-400 with DMANet and list the results in Table 6. Although Kinetics-400 does not require much temporal reasoning, DMANet achieves 78.5% which may be attributed to the reduction of redundant spatial features as well as the emphasis on key objects. Our DMANet achieves inferior performance compared to powerful methods such as SlowFast [18], Small-BigNet [70] and CorrNet [68]. However, they are built with time-consuming operations such as non-local operators or multiple streams which own heavy computations. Compared to other methods with similar efficiency such as STM [13], TEA [14] and TEINet [30], our DMANet obtains better performance. Compared to our baseline TSM [28], DMANet outperforms it by 2.8%, benefiting completely from architecture upgrade.

5 Conclusion

In this paper, based on temporal difference operators, we present a lightweight framework (DMANet) to model human dynamics toward efficient action recognition. Our DMANet is composed of a temporal difference module and a differential motion module alternatively. The temporal difference module is built up of 2D convolutions computationally free with temporal difference operators. The differential motion module guides models to pay more attention to informative regions. Our DMANet outperforms other efficient methods as well as costly 3D CNNs and two-stream methods on two temporal-related datasets and the large-scale Kinetics-400 dataset.

Author Contributions Caifeng Liu contributed to conceptualization, data curation, formal analysis, investigation, resources, validation, visualization, writing—original draft, writing—review and editing, and provided software. Fangjie Gu was involved in conceptualization, project administration, investigation, methodology, project administration, resources, validation, and writing—review and editing

Data availability statement All datasets analyzed during this study are available in links.

Declarations

Conflict of interest The authors declare no conflict of interest.

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